📊 KEY FINDINGS

🟢 Dataset Overview

- The dataset consists of **9 related tables** covering e-commerce operations including customers, orders, products, payments, and logistics.
- The data spans a **2-year period (2016–2018)**, providing sufficient coverage for trend and performance analysis.

💳 Payment Insights

- **Credit card** is the dominant payment method, accounting for **~73.92%** of all transactions.
- **Debit card** usage is minimal at **~1.47%**, indicating low adoption.

👉 **Insight:**
The platform relies heavily on credit-based transactions, suggesting customer preference for flexible payment options.

📦 Product Insights

- The platform contains **32,951 unique products.**
- Product pricing:
  - **Max price:** 6,735
  - **Min price:** 0.87
  - **Average price:** 120.65

👉 **Insight:**
The marketplace supports a **wide pricing range**, accommodating both low-cost and premium products.

🚚 Delivery Time Observations

- Delivery duration (approval → customer delivery) averages **~12 days.**
- Some orders show **unusually long delivery times**, likely due to logistics or regional factors.
- These outliers were retained as they reflect **real customer experience.**
- Delivered orders account for **~97%**, while all other statuses make up ~3%.

👉 **Insight:**
The platform demonstrates **high fulfillment success**, though delivery variability exists.

⏱️ Estimated vs Actual Delivery

- Estimated delivery time averages **~23 days.**
- Actual delivery time averages **~12 days.**
- Most estimated deliveries fall between **18–25 days**, peaking around **21 days.**

👉 **Insight:**
The company uses **conservative delivery estimates**, consistently delivering earlier than promised, which likely improves customer satisfaction.

📍 Logistics Insight

- Delivery time increases with **distance between seller and customer**, confirming that logistics performance is geographically influenced.
- A few **same-day deliveries** exist, likely due to:
  - Local fulfillment
  - Special product types
  - Timestamp inconsistencies

🧹 DATA QUALITY SUMMARY

📦 Products Table

- No duplicate product IDs found.
- **610 products** are missing catalog metadata:
  - category
  - description
  - name length
  - photo count

Cleaning Actions

- Missing categories replaced with **"unknown"**
- Missing photo counts replaced with **0**
- 2 records with missing physical attributes were retained but excluded from logistics analysis

👉 **Result:** Dataset remains complete and usable without breaking relationships.

👥 Customers Table

- No duplicate customer records
- No missing critical fields

Standardization

- City names formatted consistently
- State codes converted to uppercase
- Postal codes validated

👉 **Result:** Clean and ready for geographic analysis

📦 Orders Table

Order Status Distribution

- Delivered: **97.02%**
- Others (combined): **~3%**

Missing Values (Expected)

- Approval date: 160
- Carrier date: 1,783
- Delivery date: 2,965

👉 These are linked to **non-delivered or incomplete orders**

Time Coverage

- First order: 2016
- Last order: 2018

👉 **Result:** Reliable dataset for time-series and operational analysis

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🚚 Delivery Performance Metrics

- **Avg delivery time:** 12 days
- **Carrier handling:** 2 days
- **Estimated delivery:** 23 days

👉 **Interpretation:**

- Majority of time is spent **before shipment (processing & handling)**
- Logistics execution (carrier stage) is relatively fast

🎯 FINAL ANALYTICAL INSIGHT

- The platform is **operationally strong**, with high delivery success rates
- Delivery estimates are **intentionally conservative**, enhancing customer experience
- Logistics performance varies by **distance and region**
- Data quality is high, with minimal cleaning required
- Product and pricing structure supports a **diverse marketplace strategy**